# Analysis

April 12, 2026

## 1 Analysis

{important} AI Disclosure The written content in this report was drafted by Jenny Lee and refined using Generative AI for clarity and grammar. All ideas, analysis decisions, and interpretations are original. The prompt used throughout was: *"Refine my words."* and *Fix errors in code* when addressing issues in code written by me.

[158]:
```
import pandas as pd
import numpy as np
import requests
from concurrent.futures import ThreadPoolExecutor, as_completed
import time
import json
import ast
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import Ridge
from sklearn.preprocessing import PolynomialFeatures
from sklearn.pipeline import Pipeline
from sklearn.model_selection import cross_val_score, LeaveOneOut
from sklearn.metrics import mean_absolute_error, mean_squared_error
from sklearn.model_selection import RandomizedSearchCV, train_test_split
from scipy.stats import loguniform
import pickle, json
from itertools import combinations
from sklearn.linear_model import Lasso
from sklearn.manifold import TSNE

from sklearn.preprocessing import StandardScaler, MultiLabelBinarizer
from sklearn.compose import ColumnTransformer
from sklearn.cluster import KMeans
from sklearn.impute import SimpleImputer
from sklearn.metrics import silhouette_score
```

## 1.1 Review Data

[153]:
```
df = pd.read_csv("../data/box_office_data_with_starmeter.csv")
display(df.head())
```

```
           id   type                  primaryTitle  \
0  tt12042730  movie             Project Hail Mary
1  tt28650488  movie  The Super Mario Galaxy Movie
2  tt32430579  movie                     Crime 101
3  tt33071426  movie                     The Drama
4  tt37209937  movie                   Pizza Movie

                originalTitle  \
0            Project Hail Mary
1  The Super Mario Galaxy Movie
2                    Crime 101
3                    The Drama
4                  Pizza Movie

                                      primaryImage  startYear  \
0  {'url': 'https://m.media-amazon.com/images/M/M…     2026.0
1  {'url': 'https://m.media-amazon.com/images/M/M…     2026.0
2  {'url': 'https://m.media-amazon.com/images/M/M…     2026.0
3  {'url': 'https://m.media-amazon.com/images/M/M…     2026.0
4  {'url': 'https://m.media-amazon.com/images/M/M…     2026.0

   runtimeSeconds                                            genres  \
0          9360.0                  ['Adventure', 'Comedy', 'Sci-Fi']
1          5880.0  ['Animation', 'Adventure', 'Comedy', 'Family',…
2          8400.0                     ['Crime', 'Drama', 'Thriller']
3          6300.0                     ['Comedy', 'Drama', 'Romance']
4          5820.0                                         ['Comedy']

                                         rating  \
0  {'aggregateRating': 8.4, 'voteCount': 178625}
1   {'aggregateRating': 6.5, 'voteCount': 25522}
2   {'aggregateRating': 6.8, 'voteCount': 48321}
3   {'aggregateRating': 7.5, 'voteCount': 17072}
4    {'aggregateRating': 5.8, 'voteCount': 4258}

                                            plot  worldwideGross  \
0  A science teacher wakes up alone on a spaceshi…     433030505.0
1  Mario ventures into space, exploring cosmic wo…     437751829.0
2  An elusive thief, eyeing his final score, enco…      72559167.0
3  A happily engaged couple is put to the test wh…      26232083.0
4  High college students face an unexpectedly epi…             NaN

  worldwideGrossCurrency  productionBudget productionBudgetCurrency  \
```

```
0                    USD       200000000.0                      USD
1                    USD       110000000.0                      USD
2                    USD        90000000.0                      USD
3                    USD               NaN                      NaN
4                    NaN               NaN                      NaN

   totalStarMeter
0              37
1              66
2              49
3              27
4              25
```

### 1.2 A4

Apply an imputation to the missing observations in your dataset. Explain what you think the missingness pattern is: MCAR, MAR, or NMAR, and what this means for your imputation and future analysis.

[114]:
```python
def missing_summary(df, sort=True):
    summary = pd.DataFrame({
        'missing_count': df.isna().sum(),
        'missing_percent': round(df.isna().mean() * 100, 2)})

    if sort:
        summary = summary.sort_values(by='missing_count', ascending=False)

    return summary

print("Displaying the number of missing values in each column:")
missing_summary(df)
```

```
Displaying the number of missing values in each column:
```

[114]:

| | missing_count | missing_percent |
|---|---|---|
| productionBudget | 232 | 23.2 |
| productionBudgetCurrency | 232 | 23.2 |
| worldwideGross | 138 | 13.8 |
| worldwideGrossCurrency | 138 | 13.8 |
| rating | 52 | 5.2 |
| runtimeSeconds | 36 | 3.6 |
| startYear | 5 | 0.5 |
| primaryImage | 4 | 0.4 |
| plot | 2 | 0.2 |
| id | 0 | 0.0 |
| type | 0 | 0.0 |
| primaryTitle | 0 | 0.0 |
| originalTitle | 0 | 0.0 |

```
genres                                  0              0.0
totalStarMeter                          0              0.0
```

The dataframe above summarizes the number of missing values in each column. For the subsequent analysis, `productionBudget` and `worldwideGross` will serve as key response variables. As such, imputing missing values for these fields is avoided, since doing so could introduce significant bias and lead to unreliable model training.

The primary objective of this model is to predict the profit-to-production-budget ratio for each movie, providing producers and studios with a clearer expectation of potential financial performance. Consequently, `rating` is excluded as a feature, as it is not available prior to a movie's release and therefore would not be useful for predictive modeling in this context.

Therefore, `runtimeSeconds` is the only feature for which we will perform imputation. First, we should inspect the missing rows.

[115]:
```
df[df["runtimeSeconds"].isna()][["genres", "startYear", "runtimeSeconds"]]
```

[115]:
```
                                                genres  startYear  \
13   ['Action', 'Adventure', 'Family', 'Fantasy', '…     2026.0
33                   ['Action', 'Adventure', 'Sci-Fi']     2026.0
46       ['Action', 'Adventure', 'Fantasy', 'History']     2026.0
59        ['Action', 'Adventure', 'Fantasy', 'Sci-Fi']     2026.0
83           ['Action', 'Drama', 'Sci-Fi', 'Thriller']     2026.0
93                   ['Action', 'Adventure', 'Sci-Fi']     2026.0
124        ['Action', 'Adventure', 'Drama', 'Fantasy']     2027.0
148  ['Action', 'Adventure', 'Mystery', 'Sci-Fi', '…     2026.0
165                                          ['Drama']     2027.0
168                              ['Horror', 'Sci-Fi']     2026.0
171                              ['Comedy', 'Horror']     2026.0
224  ['Action', 'Adventure', 'Comedy', 'Family', 'F…     2026.0
230                ['Action', 'Adventure', 'Fantasy']        NaN
246                            ['Action', 'Thriller']     2026.0
254  ['Action', 'Adventure', 'Fantasy', 'Sci-Fi', '…     2026.0
314        ['Drama', 'Horror', 'Mystery', 'Thriller']        NaN
372                                         ['Comedy']     2027.0
376           ['Action', 'Crime', 'Drama', 'Mystery']     2027.0
413      ['Action', 'Adventure', 'Comedy', 'Fantasy']     2026.0
509                               ['Action', 'Drama']     2026.0
522        ['Action', 'Adventure', 'Drama', 'Fantasy']     2026.0
534      ['Action', 'Adventure', 'Fantasy', 'Sci-Fi']     2027.0
674          ['Action', 'Crime', 'Drama', 'Thriller']        NaN
683      ['Adventure', 'Comedy', 'Family', 'Mystery']     2026.0
755                            ['Adventure', 'Drama']     2026.0
775  ['Animation', 'Action', 'Adventure', 'Comedy',…     2027.0
790                              ['Action', 'Comedy']     2026.0
799                   ['Action', 'Fantasy', 'Sci-Fi']        NaN
820               ['Action', 'Adventure', 'Thriller']     2026.0
852  ['Animation', 'Adventure', 'Comedy', 'Crime', …     2026.0
```

```
856                 ['Action', 'Adventure', 'Fantasy']     2027.0
899  ['Action', 'Adventure', 'Fantasy', 'Sci-Fi', '…     2029.0
920                  ['Horror', 'Mystery', 'Thriller']     2027.0
962           ['Comedy', 'Crime', 'Drama', 'Thriller']        NaN
971                                       ['Thriller']     2027.0
994  ['Animation', 'Action', 'Adventure', 'Comedy',…     2027.0

     runtimeSeconds
13              NaN
33              NaN
46              NaN
59              NaN
83              NaN
93              NaN
124             NaN
148             NaN
165             NaN
168             NaN
171             NaN
224             NaN
230             NaN
246             NaN
254             NaN
314             NaN
372             NaN
376             NaN
413             NaN
509             NaN
522             NaN
534             NaN
674             NaN
683             NaN
755             NaN
775             NaN
790             NaN
799             NaN
820             NaN
852             NaN
856             NaN
899             NaN
920             NaN
962             NaN
971             NaN
994             NaN
```

### 1.2.1 Type of Missing Values

Upon inspecting the rows with missing `runtimeSeconds`, we observe that many of these entries have a `startYear` beyond 2026, with some films scheduled for release as late as 2027. This suggests that the missing runtime values may be due to the information not yet being publicly available. However, this remains a hypothesis and cannot be confirmed from the data alone.

Since the missingness appears to depend on information that is not captured by the observed variables (i.e., whether the runtime has been publicly released), it is reasonable to classify this as **Not Missing at Random (NMAR)**. In such cases, standard imputation methods may introduce bias.

Nevertheless, to further explore the data, we proceed with model-based imputation for `runtimeSeconds`.‘ It is important to note, however, that due to the NMAR nature of the missingness, the imputed values should be treated with caution and may not be reliable for downstream modeling.

[116]:
```
y_impute = df[df["runtimeSeconds"].isna()]
print(len(y_impute))
```

```
36
```

[125]:
```
df["genres_list"] = df["genres"].apply(lambda x: ast.literal_eval(x) if pd.
 ↪notna(x) else [])

ALL_GENRES = sorted({g for gl in df["genres_list"] for g in gl})

def genres_to_vector(genres_list, all_genres=ALL_GENRES):
    return [int(g in genres_list) for g in all_genres]

df_ = df[df["runtimeSeconds"].notna()]
X = np.array([genres_to_vector(gl) for gl in df_["genres_list"]])
y = df_["runtimeSeconds"].values

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,␣
 ↪random_state=42)

genre_df = pd.DataFrame(X, columns=ALL_GENRES)
genre_df
```

[125]:
```
     Action  Adventure  Animation  Biography  Comedy  Crime  Documentary  \
0         0          1          0          0       1      0            0
1         0          1          1          0       1      0            0
2         0          0          0          0       0      1            0
3         0          0          0          0       1      0            0
4         0          0          0          0       1      0            0
..      …        …        …        …     …    …          …
959       0          0          0          0       1      0            0
960       0          0          0          0       0      0            0
```

```
961       0          0          0          1       0      0            0
962       0          0          0          0       0      1            0
963       0          0          0          0       0      0            0

     Drama  Family  Fantasy  …  Horror  Music  Musical  Mystery  Romance  \
0        0       0        0  …       0      0        0        0        0
1        0       1        1  …       0      0        0        0        0
2        1       0        0  …       0      0        0        0        0
3        1       0        0  …       0      0        0        0        1
4        0       0        0  …       0      0        0        0        0
..     …      …       …  …     …    …       …       …       …
959      0       0        0  …       0      0        0        0        0
960      1       0        0  …       0      0        0        0        1
961      1       0        0  …       0      0        0        0        0
962      1       0        0  …       0      0        0        0        1
963      1       0        0  …       1      0        0        0        0

     Sci-Fi  Sport  Thriller  War  Western
0         1      0         0    0        0
1         0      0         0    0        0
2         0      0         1    0        0
3         0      0         0    0        0
4         0      0         0    0        0
..      …    …       …  …      …
959       0      0         0    0        0
960       0      0         0    0        0
961       0      0         0    0        0
962       0      0         1    0        0
963       0      0         1    0        0

[964 rows x 21 columns]
```

### 1.2.2 Data Transformation

To estimate the missing values of `runtimeSeconds`, we use related features such as the movie's genres. Since genres are provided as a list (with each movie potentially belonging to multiple genres), the column must first be transformed into a binary (one-hot encoded) format. This allows each genre to be represented as an individual feature, which can then be used as input for building the imputation model.

[119]:
```
genre_stats = {}
genre_rows = []
for g in ALL_GENRES:
    mask = genre_df[g] == 1
    if mask.sum() > 0:
        vals = y[mask]
        mean_rt = vals.mean()
```

```
        genre_stats[g] = {
            "count": int(mask.sum()),
            "mean_seconds": round(mean_rt, 1),
            "mean_minutes": round(mean_rt / 60, 1),
        }
        genre_rows.append({
            "genre": g,
            "count": int(mask.sum()),
            "mean_min": round(mean_rt / 60, 1)})

genre_stats_df = (pd.DataFrame(genre_rows).sort_values("mean_min",␣
 ↪ascending=False).reset_index(drop=True))
genre_stats_df
```

[119]:

| | genre | count | mean_min |
|---|---|---|---|
| 0 | Western | 15 | 2.4 |
| 1 | Biography | 61 | 2.3 |
| 2 | War | 54 | 2.2 |
| 3 | Sport | 19 | 2.1 |
| 4 | Music | 27 | 2.0 |
| 5 | Romance | 133 | 2.0 |
| 6 | Documentary | 7 | 1.7 |
| 7 | Action | 333 | NaN |
| 8 | Adventure | 275 | NaN |
| 9 | Animation | 59 | NaN |
| 10 | Comedy | 279 | NaN |
| 11 | Crime | 181 | NaN |
| 12 | Drama | 526 | NaN |
| 13 | Family | 94 | NaN |
| 14 | Fantasy | 152 | NaN |
| 15 | History | 54 | NaN |
| 16 | Horror | 147 | NaN |
| 17 | Musical | 29 | NaN |
| 18 | Mystery | 167 | NaN |
| 19 | Sci-Fi | 199 | NaN |
| 20 | Thriller | 361 | NaN |

### 1.2.3 Model-Based Imputation

[127]:

```
pipeline = Pipeline([("interactions", PolynomialFeatures(degree=3,␣
 ↪interaction_only=True)),
                     ("lasso", Lasso(max_iter=10000))])

pipeline.fit(X, y)

search = RandomizedSearchCV(
    pipeline.fit(X, y),
```

```
    param_distributions={"lasso__alpha": loguniform(0.01, 10_000)},
    n_iter=100,
    scoring="neg_root_mean_squared_error",
    cv=10,
    random_state=123,
    n_jobs=-1,
)
search.fit(X_train, y_train)
```

```
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 4.750e+06, tolerance: 1.693e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.825e+06, tolerance: 1.552e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 2.180e+07, tolerance: 1.632e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 9.328e+06, tolerance: 1.702e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 4.620e+05, tolerance: 1.733e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 7.285e+06, tolerance: 1.680e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
```

```
check the scale of the features or consider increasing regularisation. Duality
gap: 5.423e+06, tolerance: 1.662e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.914e+06, tolerance: 1.665e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.269e+06, tolerance: 1.620e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 7.991e+05, tolerance: 1.680e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 2.687e+06, tolerance: 1.632e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 8.808e+05, tolerance: 1.614e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 4.278e+05, tolerance: 1.665e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.038e+06, tolerance: 1.552e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
```

```
check the scale of the features or consider increasing regularisation. Duality
gap: 6.613e+05, tolerance: 1.702e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 4.336e+05, tolerance: 1.620e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.056e+06, tolerance: 1.632e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.918e+05, tolerance: 1.552e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 5.855e+05, tolerance: 1.614e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 4.809e+05, tolerance: 1.733e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 6.343e+05, tolerance: 1.693e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 7.386e+05, tolerance: 1.632e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
```

```
check the scale of the features or consider increasing regularisation. Duality
gap: 3.787e+05, tolerance: 1.614e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.730e+05, tolerance: 1.620e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 8.007e+05, tolerance: 1.693e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.221e+06, tolerance: 1.614e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 3.118e+06, tolerance: 1.632e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.736e+06, tolerance: 1.552e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.699e+06, tolerance: 1.702e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 9.428e+05, tolerance: 1.693e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
```

```
check the scale of the features or consider increasing regularisation. Duality
gap: 1.454e+06, tolerance: 1.680e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.615e+06, tolerance: 1.665e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 8.510e+05, tolerance: 1.552e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.992e+06, tolerance: 1.632e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 9.255e+05, tolerance: 1.614e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 4.907e+05, tolerance: 1.680e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 3.496e+05, tolerance: 1.620e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 4.899e+05, tolerance: 1.665e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
```

```
check the scale of the features or consider increasing regularisation. Duality
gap: 2.167e+05, tolerance: 1.702e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 4.742e+07, tolerance: 1.614e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.909e+07, tolerance: 1.552e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 3.609e+07, tolerance: 1.620e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 4.886e+07, tolerance: 1.632e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 5.103e+07, tolerance: 1.680e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.141e+07, tolerance: 1.665e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 3.450e+07, tolerance: 1.662e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
```

```
check the scale of the features or consider increasing regularisation. Duality
gap: 4.573e+07, tolerance: 1.702e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 7.837e+06, tolerance: 1.733e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 3.407e+07, tolerance: 1.693e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 4.408e+05, tolerance: 1.662e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 4.555e+06, tolerance: 1.614e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 3.003e+07, tolerance: 1.680e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 8.222e+06, tolerance: 1.552e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 6.509e+06, tolerance: 1.620e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
```

```
check the scale of the features or consider increasing regularisation. Duality
gap: 2.363e+07, tolerance: 1.632e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.462e+07, tolerance: 1.702e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.650e+07, tolerance: 1.665e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 1.802e+06, tolerance: 1.733e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 2.850e+07, tolerance: 1.662e+05
  model = cd_fast.enet_coordinate_descent(
/Users/jennylee/miniforge3/envs/ds/lib/python3.13/site-
packages/sklearn/linear_model/_coordinate_descent.py:695: ConvergenceWarning:
Objective did not converge. You might want to increase the number of iterations,
check the scale of the features or consider increasing regularisation. Duality
gap: 4.820e+06, tolerance: 1.693e+05
  model = cd_fast.enet_coordinate_descent(
```

[127]:
```
RandomizedSearchCV(cv=10,
                   estimator=Pipeline(steps=[('interactions',
                                              PolynomialFeatures(degree=3,
interaction_only=True)),
                                             ('lasso', Lasso(max_iter=10000))]),
                   n_iter=100, n_jobs=-1,
                   param_distributions={'lasso__alpha':
<scipy.stats._distn_infrastructure.rv_continuous_frozen object at 0x31f2fe2c0>},
                   random_state=123, scoring='neg_root_mean_squared_error')
```

To build the imputation model, we applied the `PolynomialFeatures` preprocessing tool to capture interaction effects between genres, recognizing that combinations of genres may influence runtime beyond their individual contributions. To mitigate overfitting, the interaction degree was limited to 3, allowing up to three-way interactions among genres.

For the modeling step, **Lasso regression** was used due to its inherent ability to perform feature selection and reduce dimensionality, which is particularly important given the large number of interaction terms generated. Model performance is evaluated using **Root Mean Squared Error (RMSE)**, as it provides an interpretable measure of prediction error in the same units as the target variable.

The hyperparameter, $\alpha$, is tuned using `RandomizedSearchCV`, which repeatedly samples values from a predefined distribution and evaluates model performance through cross-validation to identify the optimal level of regularization.

[130]:
```
cv_results = (
    pd.DataFrame(search.cv_results_)[["param_lasso__alpha", "mean_test_score"]]
    .rename(columns={"param_lasso__alpha": "alpha", "mean_test_score":␣
 ↪"neg_rmse"})
    .head(5)
    .reset_index(drop=True)
)
cv_results
```

[130]:

| | alpha | neg_rmse |
|---|---|---|
| 0 | 150.943742 | -1498.872655 |
| 1 | 0.520998 | -1533.139475 |
| 2 | 0.229672 | -1633.554978 |
| 3 | 20.318358 | -1346.933094 |
| 4 | 207.402420 | -1532.825140 |

[132]:
```
best_alpha = search.best_params_["lasso__alpha"]
print(f"Best alpha: {best_alpha}")
```

```
Best alpha: 10.256992909752547
```

[133]:
```
pipeline  = search.best_estimator_
y_pred    = pipeline.predict(X_test)
rmse_test = np.sqrt(mean_squared_error(y_test, y_pred))
mae_test  = mean_absolute_error(y_test, y_pred)

print(f"Test RMSE: {rmse_test:.2f} seconds ({rmse_test/60:.2f} minutes)")
print(f"Test MAE: {mae_test:.2f} seconds ({mae_test/60:.2f} minutes)")
```

```
Test RMSE: 1306.54 seconds (21.78 minutes)
Test MAE: 1013.89 seconds (16.90 minutes)
```

[136]:
```
X_pred = np.array([genres_to_vector(gl) for gl in y_impute["genres"]])
y_pred = pipeline.predict(X_pred)
```

[137]:
```
y_impute["runtimeSeconds"] = y_pred
df_comb = pd.concat([df[~df["runtimeSeconds"].isna()], y_impute],␣
 ↪ignore_index=True)
```

```
/var/folders/c5/pphx1c0n3lb12sy85_ybqxqm0000gn/T/ipykernel_21652/603644274.py:1:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  y_impute["runtimeSeconds"] = y_pred
```

[138]:
```
missing_summary(df_comb)
```

[138]:

| | missing_count | missing_percent |
|---|---|---|
| productionBudget | 232 | 23.2 |
| productionBudgetCurrency | 232 | 23.2 |
| worldwideGross | 138 | 13.8 |
| worldwideGrossCurrency | 138 | 13.8 |
| rating | 52 | 5.2 |
| genres_list | 36 | 3.6 |
| startYear | 5 | 0.5 |
| primaryImage | 4 | 0.4 |
| plot | 2 | 0.2 |
| id | 0 | 0.0 |
| type | 0 | 0.0 |
| primaryTitle | 0 | 0.0 |
| originalTitle | 0 | 0.0 |
| runtimeSeconds | 0 | 0.0 |
| genres | 0 | 0.0 |
| totalStarMeter | 0 | 0.0 |

We now observe that there are no remaining missing values in `runtimeSeconds`.

However, entries with missing values in `productionBudget` and `worldwideGross` must be removed, as these variables serve as the response variables in our model. Notably, all observations that originally had missing `runtimeSeconds` also lack values for these two variables. As a result, the imputed `runtimeSeconds` values are ultimately excluded from the modeling dataset, which avoids introducing potentially unreliable imputed data into the analysis.

[139]:
```
df = df[~df["productionBudget"].isna() & df["worldwideGross"].notna()]
print(f"After removing rows with missing productionBudget and worldwideGross,␣
↪we have {len(df)} rows left.")
```

```
After removing rows with missing productionBudget and worldwideGross, we have
743 rows left.
```

### 1.3 A1

Write about any necessary transformations to the values of the data. This is different from E2, which asks about transformations to the format of the data. For example, if your model is producing non-normal residuals, and but it relies on normally distributed residuals, then talk about the log-transform or Box-Cox transform necessary for this.

### 1.3.1 Data Transformation

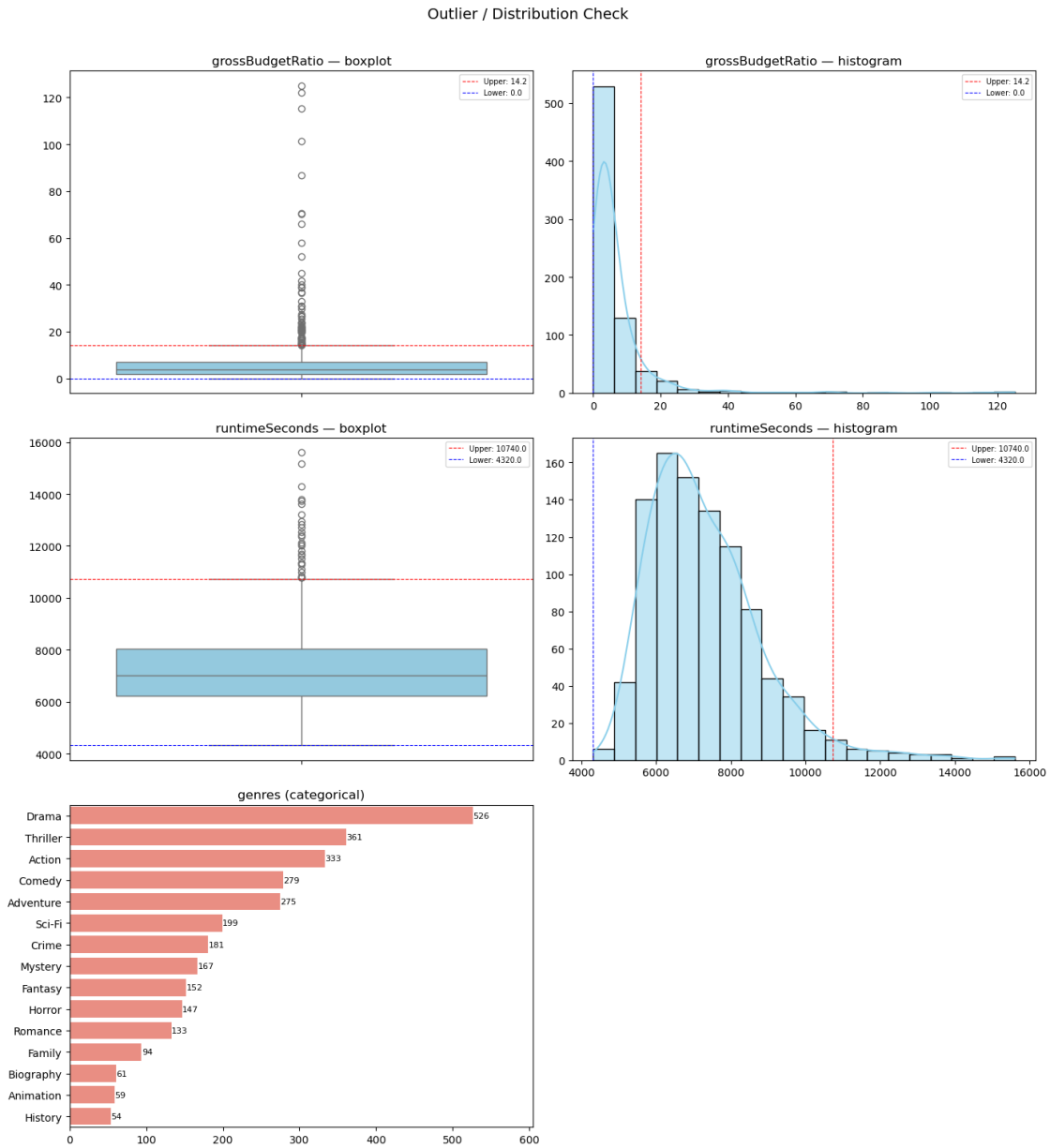
Before examining potential outliers, we first engineer a new feature, `grossBudgetRatio`, defined as the ratio of `worldwideGross` to `productionBudget`. This variable will serve as the response for subsequent clustering analysis.

[154]:
```
df["grossBudgetRatio"] = df["worldwideGross"] / df["productionBudget"]
```

### 1.3.2 Identifying Outliers

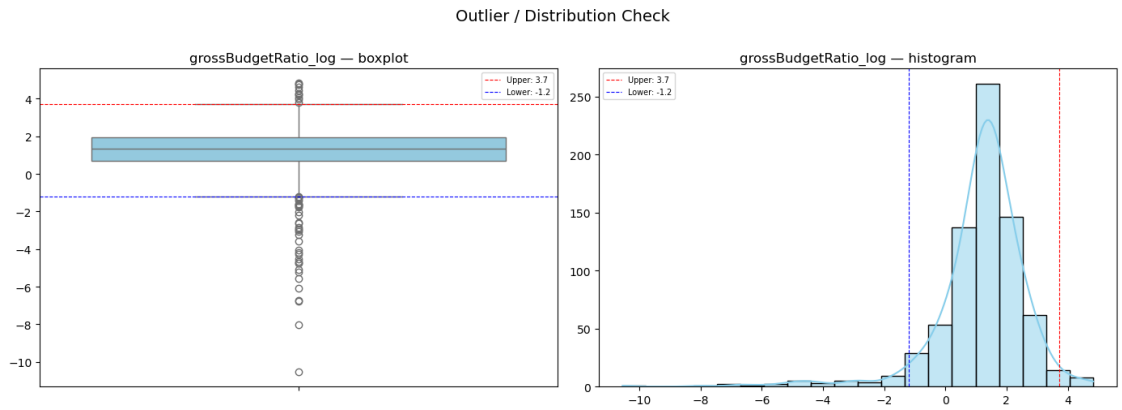
[155]:
```
def plot_outliers(df, columns, ncols=2):
    numeric_cols = [c for c in columns if c in df.columns and pd.api.types.
↪is_numeric_dtype(df[c])]
    cat_cols     = [c for c in columns if c in df.columns and not pd.api.types.
↪is_numeric_dtype(df[c])]

    cat_rows   = max(1, -(-len(cat_cols) // ncols)) if cat_cols else 0
    total_rows = len(numeric_cols) + cat_rows

    fig, axes = plt.subplots(total_rows, ncols, figsize=(14, 5 * total_rows))
    axes = np.array(axes).reshape(total_rows, ncols)

    for row_idx, col in enumerate(numeric_cols):
        data = df[col].dropna()
        Q1, Q3 = data.quantile(0.25), data.quantile(0.75)
        IQR    = Q3 - Q1
        actual_upper = data[data <= Q3 + 1.5 * IQR].max()
        actual_lower = data[data >= Q1 - 1.5 * IQR].min()

        ax_box = axes[row_idx, 0]
        sns.boxplot(y=data, ax=ax_box, color="skyblue")
        ax_box.axhline(actual_upper, color="red",  linestyle="--", linewidth=0.
↪8, label=f"Upper: {actual_upper:.1f}")
        ax_box.axhline(actual_lower, color="blue", linestyle="--", linewidth=0.
↪8, label=f"Lower: {actual_lower:.1f}")
        ax_box.legend(fontsize=7)
        ax_box.set_title(f"{col} - boxplot")
        ax_box.set_xlabel(""); ax_box.set_ylabel("")

        ax_hist = axes[row_idx, 1]
        sns.histplot(data, ax=ax_hist, color="skyblue", kde=True, bins=20)
        ax_hist.axvline(actual_upper, color="red",  linestyle="--", linewidth=0.
↪8, label=f"Upper: {actual_upper:.1f}")
        ax_hist.axvline(actual_lower, color="blue", linestyle="--", linewidth=0.
↪8, label=f"Lower: {actual_lower:.1f}")
        ax_hist.legend(fontsize=7)
        ax_hist.set_title(f"{col} - histogram")
```

```
        ax_hist.set_xlabel(""); ax_hist.set_ylabel("")

    for cat_idx, col in enumerate(cat_cols):
        row_idx = len(numeric_cols) + cat_idx // ncols
        col_idx = cat_idx % ncols
        ax = axes[row_idx, col_idx]

        series = df[col].dropna()
        def parse_entry(x):
            if isinstance(x, list): return x
            if isinstance(x, str) and x.startswith("["):
                try: return ast.literal_eval(x)
                except: return [x]
            return [x]
        series = series.apply(parse_entry).explode().str.strip().str.
↪strip("'\"")
        counts = series.value_counts().head(15)

        bars = sns.barplot(x=counts.values, y=counts.index, ax=ax,␣
↪color="salmon")
        for bar, count in zip(bars.patches, counts.values):
            ax.text(bar.get_width() + 0.3, bar.get_y() + bar.get_height() / 2,
                    str(count), va="center", ha="left", fontsize=8)
        ax.set_xlim(0, counts.values[0] * 1.15)
        ax.set_title(f"{col} (categorical)")
        ax.set_xlabel(""); ax.set_ylabel("")

    for j in range(len(cat_cols), cat_rows * ncols):
        axes[len(numeric_cols) + j // ncols, j % ncols].set_visible(False)

    plt.suptitle("Outlier / Distribution Check", fontsize=14, y=1.01)
    plt.tight_layout()
    plt.show()

plot_outliers(df, ["grossBudgetRatio", "runtimeSeconds", "genres"])
```




From the subplots above, it is evident that `grossBudgetRatio` is severely skewed. Since this variable will be used as the response in a regression model, a **log transformation** should be applied to improve its distribution. This transformation reduces the influence of extreme values, allowing the model to treat observations more evenly, and helps stabilize the variance, leading to more reliable modeling.

[156]:
```
df["grossBudgetRatio_log"] = np.log(df["grossBudgetRatio"])
plot_outliers(df, ["grossBudgetRatio_log"])
```




We now observe that the `grossBudgetRatio_log` feature exhibits a more evenly distributed range of values.

## 1.4 A3

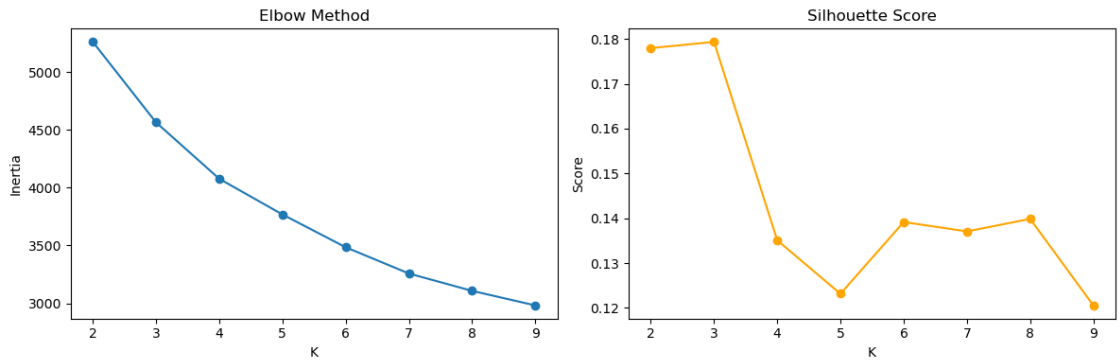
Apply a clustering method to find groups of observations. This can be a K-means clustering, or a t-SNE.

[160]:
```
def parse_rating(x):
    try:
        d = ast.literal_eval(x)
        return d.get("aggregateRating", np.nan), d.get("voteCount", np.nan)
    except:
        return np.nan, np.nan

def parse_genres(x):
    try:
        return ast.literal_eval(x) if isinstance(x, str) else []
    except:
        return []

df["aggregateRating"], df["voteCount"] = zip(*df["rating"].apply(parse_rating))
df["genres_parsed"] = df["genres"].apply(parse_genres)

mlb = MultiLabelBinarizer()
genre_encoded = pd.DataFrame(
    mlb.fit_transform(df["genres_parsed"]),
    columns=[f"genre_{g}" for g in mlb.classes_],
    index=df.index
)
display(genre_encoded.head())
```

```
numeric_features = ["runtimeSeconds", "worldwideGross", "productionBudget",␣
↪"totalStarMeter", "aggregateRating", "voteCount"]

model_df = pd.concat([df[numeric_features], genre_encoded], axis=1).dropna()

genre_features = list(genre_encoded.columns)

preprocessor = ColumnTransformer(transformers=[
    ("num", Pipeline([
        ("imputer", SimpleImputer(strategy="median")),
        ("scaler", StandardScaler())
    ]), numeric_features),
    ("genre", "passthrough", genre_features)
])

inertias = []
silhouettes = []
K_range = range(2, 10)

X = preprocessor.fit_transform(model_df)

for k in K_range:
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = km.fit_predict(X)
    inertias.append(km.inertia_)
    silhouettes.append(silhouette_score(X, labels))

fig, axes = plt.subplots(1, 2, figsize=(12, 4))
axes[0].plot(K_range, inertias, marker="o")
axes[0].set_title("Elbow Method")
axes[0].set_xlabel("K")
axes[0].set_ylabel("Inertia")

axes[1].plot(K_range, silhouettes, marker="o", color="orange")
axes[1].set_title("Silhouette Score")
axes[1].set_xlabel("K")
axes[1].set_ylabel("Score")

plt.tight_layout()
plt.show()

best_k = silhouettes.index(max(silhouettes)) + 2
print(f"Best K by silhouette: {best_k}")

pipeline = Pipeline([
    ("preprocessor", preprocessor),
    ("kmeans", KMeans(n_clusters=best_k, random_state=42, n_init=10))
```

```
])

pipeline.fit(model_df)
df.loc[model_df.index, "cluster"] = pipeline.named_steps["kmeans"].labels_
df["cluster"] = df["cluster"].astype("Int64")

summary = df.groupby("cluster")[numeric_features].mean().round(2)
display(summary)
```

```
   genre_Action  genre_Adventure  genre_Animation  genre_Biography  \
0             0                1                0                0
1             0                1                1                0
2             0                0                0                0
3             0                0                0                0
4             0                0                0                0

   genre_Comedy  genre_Crime  genre_Documentary  genre_Drama  genre_Family  \
0             1            0                  0            0             0
1             1            0                  0            0             1
2             0            1                  0            1             0
3             1            0                  0            1             0
4             1            0                  0            0             0

   genre_Fantasy  …  genre_Horror  genre_Music  genre_Musical  \
0              0  …             0            0              0
1              1  …             0            0              0
2              0  …             0            0              0
3              0  …             0            0              0
4              0  …             0            0              0

   genre_Mystery  genre_Romance  genre_Sci-Fi  genre_Sport  genre_Thriller  \
0              0              0             1            0               0
1              0              0             0            0               0
2              0              0             0            0               1
3              0              1             0            0               0
4              0              0             0            0               0

   genre_War  genre_Western
0          0              0
1          0              0
2          0              0
3          0              0
4          0              0

[5 rows x 21 columns]
```




```
Best K by silhouette: 3

         runtimeSeconds  worldwideGross  productionBudget  totalStarMeter  \
cluster
0               8439.50    5.100839e+08      1.115219e+08           15.36
1               6895.56    1.544829e+08      5.056319e+07           12.01
2               7580.00    1.205292e+08      1.166667e+10            0.00

         aggregateRating  voteCount
cluster
0                   7.94  893291.77
1                   6.97  272944.86
2                   8.10  439095.67
```